

INTERSHIP REPORT

Artificial Intelligence Internship

Codec Technologies Pvt. Ltd.

B.Tech CSE • Batch 2024–2028

SACHIN PANDEY

Roll No. 2410031664

Internship Period: 11 June 2026 – 11 July 2026

AI • MACHINE LEARNING • PYTHON • IRIS CLASSIFICATION



Candidate Profile

Academic and internship details

**SACHIN
PANDEY**

B.Tech – Computer Science
& Engineering

Roll No. 2410031664

Batch 2024–2028

1

Institute

IILM University, Greater Noida, U.P.

2

Internship

1-Month Artificial Intelligence Internship

3

Organization

Codec Technologies Pvt. Ltd.

4

Approval

AICTE & ICAC Approved

5

Mode

Remote / Online

6

Role

Artificial Intelligence Intern

From AI Foundations to a Practical Project

The report describes a one-month AICTE & ICAC approved internship.

01 Foundations

Introduction to Artificial Intelligence, Machine Learning and Deep Learning concepts.

Python basics and data handling with NumPy and Pandas.

02 Core Machine Learning

Covered regression, classification and model evaluation concepts.

The learning progressed from concepts to practical use.

11/06/2026 → 11/07/2026

03 Applied Project

Final project: Iris Flower Classification.

K-Nearest Neighbors (KNN) was used to classify flower species from measurements.

Codec Technologies Pvt. Ltd.

Training organization offering short approved internship programs.

- Offers internship programs in Artificial Intelligence, Python, Data Analytics and Cloud Computing.
- The internship program was conducted online / remotely.
- The report states that the programs are AICTE & ICAC approved.
- The organization is partnered with the National Internship Portal (Ministry of Education).

INTERNSHIP SNAPSHOT

Duration **1 Month**

Role **AI Intern**

Program Manager

Dr. Anurag Shrivastava

Mode **Remote / Online**

Problem Statement & Objectives

The project focuses on classifying flower species from four numerical measurements.

Problem Statement

It is difficult for a beginner—or a machine without a learned pattern—to identify the exact flower species by appearance alone.

Using sepal length, sepal width, petal length and petal width, the model learns to distinguish three species: Setosa, Versicolor and Virginica.

PROJECT OBJECTIVES

- Understand basic AI and Machine Learning concepts practically.
- Use the Iris dataset to train a simple model.
- Classify flowers based on their measurements.
- Check how accurate the model's predictions are.
- Gain hands-on experience with Python AI tools.

Scope, Technologies & Tools

A deliberately simple project matching the level of a one-month beginner internship.

1 Python 3

Programming Language

2 Pandas

Data Handling

3 NumPy

Data Handling

4 Scikit-learn

Machine Learning

5 Matplotlib

Visualization

**6 Jupyter /
Colab**

Development

7 Iris Dataset

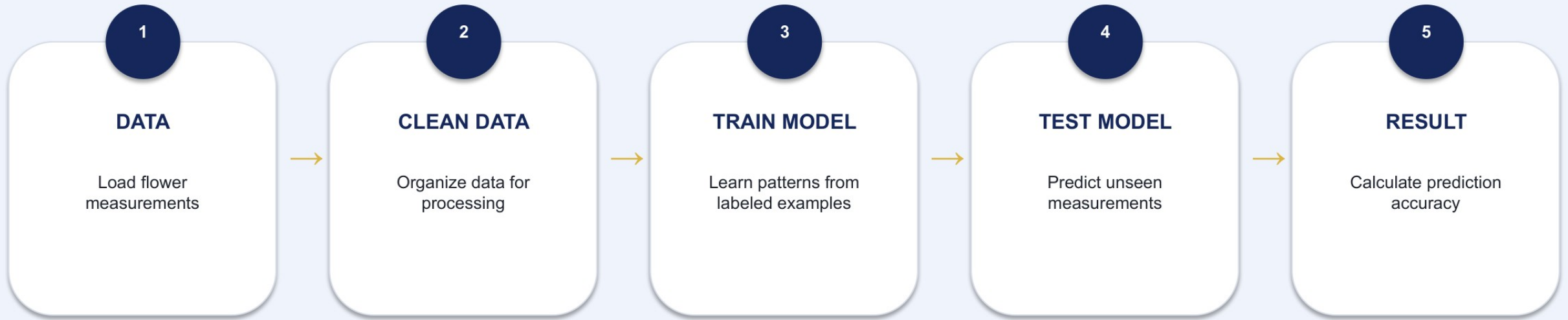
Dataset

SCOPE

Numbers, not images
Beginner-scale Iris dataset
Not intended as a large real-world product

Data → Clean → Train → Test → Result

The report presents a simple five-step workflow.



KNN is the final classification algorithm used in the project.

Week-wise Internship Learning Journey

Conceptual learning, hands-on practice and a final applied project.

1

Foundations of AI & Python

AI, ML, DL concepts; Python basics; NumPy and Pandas.

WEEK 1

2

Core Machine Learning

Regression, classification and model evaluation.

WEEK 2

3

Deep Learning Basics

Introduction to Deep Learning models and neural networks.

WEEK 3

4

Final Project

Built and tested Iris Flower Classification using KNN.

WEEK 4

Iris Flower Classification with KNN

The report describes the complete model-building sequence.

1 LOAD

150 samples
Each sample has 4 measurements and a flower label.

2 SPLIT

80% / 20%
Most data for training; the remainder for testing.

3 MODEL

KNN
Find nearby known flowers and choose the closest species.

4 TRAIN

Learn pattern
Train the model on the training data.

5 TEST

Accuracy
Evaluate predictions on unseen data.

Species: Setosa • Versicolor • Virginica

Skills & Learning Outcomes

The report states the expected results rather than presenting a measured final accuracy.

1 Working Classifier

A simple program able to guess flower species with good accuracy.

2 ML Understanding

Basic understanding of how a model is trained and tested.

3 Practical Tools

Hands-on practice with Python, Pandas and Scikit-learn.

4 AI Concepts

A clear beginner example for understanding core AI/ML ideas.

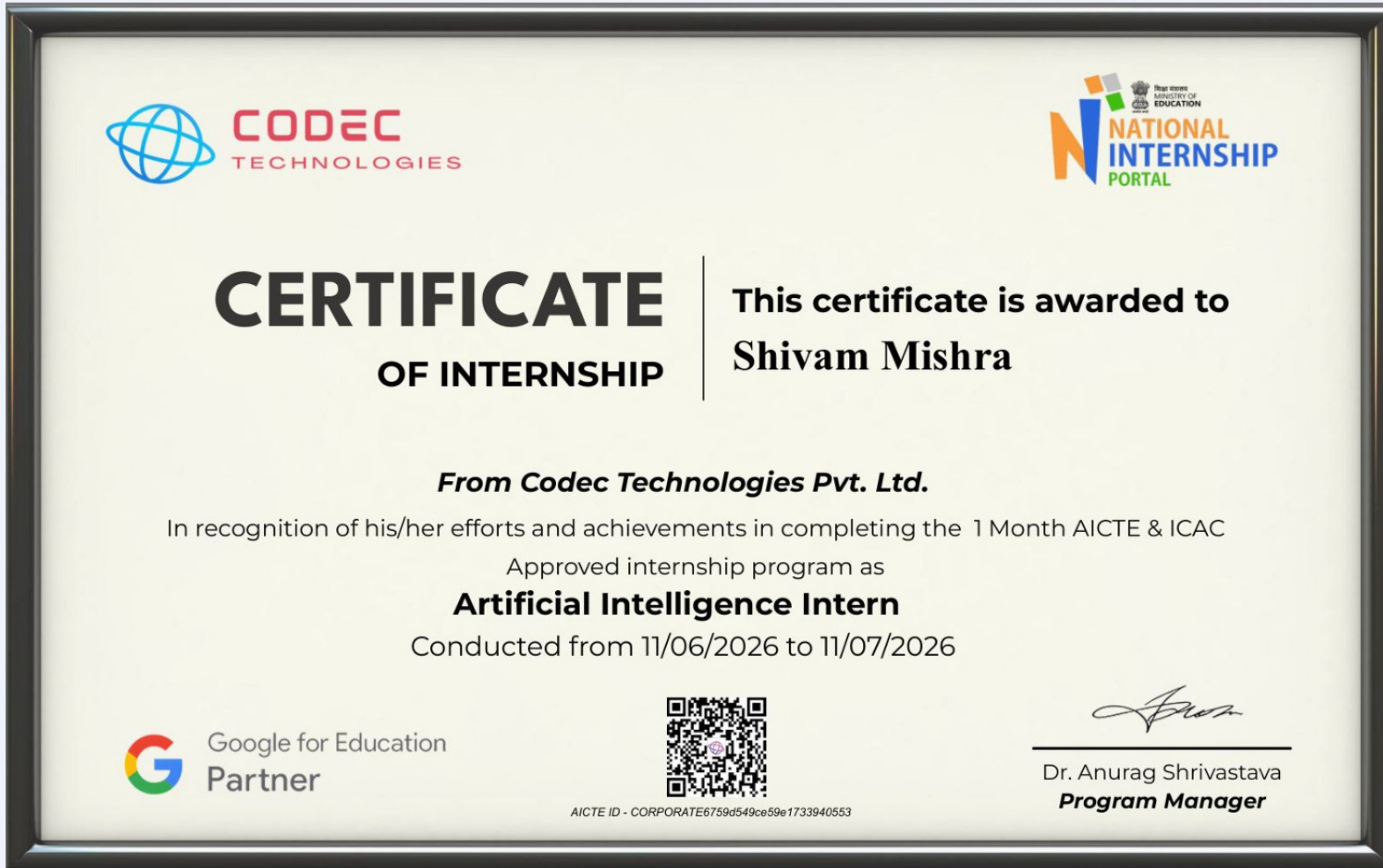
EXPECTED ACCURACY

Usually 90%+ for this dataset

This is the report's expected benchmark, not a claimed measured result.

Internship Completion Certificate

Issued by Codec Technologies Pvt. Ltd. for successful completion of the internship.



Bibliography & Source Material

References listed in the internship report.

- Codec Technologies Pvt. Ltd. – 1-Month Artificial Intelligence Internship Program.
- Scikit-learn Documentation – Machine Learning in Python.
- Pandas Documentation – Python Data Analysis Library.
- NumPy Documentation – Numerical Computing in Python.
- Matplotlib Documentation – Visualization with Python.
- Fisher, R.A. (1936) – The Use of Multiple Measurements in Taxonomic Problems.
- GeeksforGeeks – Machine Learning Tutorials and K-Nearest Neighbors.
- National Internship Portal, Ministry of Education, Government of India.

Source URLs are reproduced from the report's bibliography.

THANK YOU

Artificial Intelligence Internship

Codec Technologies Pvt. Ltd.

SACHIN PANDEY
Roll No. 2410031664

B.Tech CSE • Batch 2024–2028

11 June 2026 – 11 July 2026

